Continuous Fubini Factorization and Quadrature Architecture for the Polyatomic Spectral Boltzmann Collision Operator

1 Continuous Formulation and Spatial Kinematic Decoupling

The Wigner–Eckart factorization reduces the macroscopic Boltzmann collision operator to a rotationally invariant kinematic core. For a polyatomic gas, this core is 9-dimensional. However, the five macroscopic spatial variables can be decoupled similarly to the monatomic case.

Evaluating the integral directly in target and incident speeds (v, w) introduces a physical singularity $|v - w|^\gamma$ that is non-analytic across the $v = w$ diagonal. To decouple the kinematics and absorb this singularity, we rotate the radial domain by 45 degrees, substituting the original speeds with center-of-mass (t) and relative (s) coordinates:

$$t = \frac{v + w}{\sqrt{2}}, \quad s = \frac{v - w}{\sqrt{2}}. \tag{1}$$

To map these coordinates to spectral integration domains, we apply a square-root mapping to the energy variable $z = t^2$, and parameterize the relative sub-domain via a fractional coordinate $x = s/t \in [0, 1]$. The pre-collision speeds are reconstructed as functions of the quadrature variables:

$$v(z, x) = \sqrt{\frac{z}{2}}(1 + x), \tag{2}$$

$$w(z, x) = \sqrt{\frac{z}{2}}(1 - x). \tag{3}$$

During an elastic collision, the relative velocity vector rotates. This rotation is parameterized by the polar deflection angle $\chi \in [0, \pi]$ and the azimuthal scattering angle $\epsilon \in [0, 2\pi]$. Utilizing a non-linear half-angle transformation $y = \sin(\beta/2) \in [0, 1]$ for the incidence angle β, the pre-collision relative speed evaluates to:

$$u(z, x, y) = \sqrt{(v - w)^2 + 4vwy^2} = \sqrt{2z}\sqrt{x^2 + (1 - x^2)y^2}. \tag{4}$$

To resolve the 2D cone singularity $\sqrt{x^2 + y^2}$ at the origin, we apply a 2D Duffy Transformation, splitting the (x, y) unit square into two triangular patches mapped by an inner coordinate $h \in [0, 1]$:

- **Patch 1 ($x > y$):** Map $y = xh$. The relative speed becomes $u(z, x, h) = \sqrt{2z} \cdot x\sqrt{1 + (1 - x^2)h^2}$.
- **Patch 2 ($y > x$):** Map $x = yh$. The relative speed becomes $u(z, y, h) = \sqrt{2z} \cdot y\sqrt{h^2 + 1 - y^2h^2}$.

2 Polyatomic Collision Kinematics and Dependencies

With the 5D spatial geometry $(z, x, y, \chi, \epsilon)$ mapped, we introduce the 4D internal energy integration variables: the pre-collision internal energies $I, J \in [0, \infty)$, the kinetic partition parameter $R \in [0, 1]$, and the internal partition parameter $r \in [0, 1]$.

The functional dependencies of the polyatomic collision mechanics dictate the integration ordering. We track these dependencies as follows:

$$\text{Total Energy:} \quad E(z,x,y,I,J) = \frac{1}{2}u(z,x,y)^2 + I + J, \tag{5}$$

$$\text{Post-Collision Relative Speed:} \quad u'(z,x,y,I,J,R) = \sqrt{2RE(\dots)}, \tag{6}$$

$$\text{Post-Collision Target Internal:} \quad I'(z,x,y,I,J,R,r) = r(1-R)E(\dots). \tag{7}$$

The post-collision target velocity vector $\mathbf{v}'$ requires the pre-collision spatial geometry (which dictates the center-of-mass velocity $\mathbf{U}$), the post-collision relative speed u', and the scattering direction unit vector $\hat{\mathbf{u}}'(\chi,\epsilon)$:

$$\mathbf{v}' = \mathbf{U} + \frac{1}{2}u'\hat{\mathbf{u}}'. \tag{8}$$

From this, its magnitude v' and unit direction $\hat{\mathbf{v}}'$ are determined:

$$(v')^2 = |\mathbf{U}|^2 + \frac{1}{4}(u')^2 + (\mathbf{U}\cdot\hat{\mathbf{u}}')u', \tag{9}$$

$$\hat{\mathbf{v}}' = \frac{\mathbf{v}'}{v'}. \tag{10}$$

Both v' and $\hat{\mathbf{v}}'$ depend on the kinetic partition R, but are independent of the internal partition parameter r. Finally, the physical collision kernel evaluates the transition probability:

$$B := B(u(z,x,y), \cos\chi, I, J, R, r). \tag{11}$$

3 Asymmetric Weak Form and Continuous Sum-Factorization

Before discretizing the polyatomic manifold, we isolate the continuous scattering kinematics using the asymmetric weak form of the collision operator. Tracking only the variation of the target particle against a test function $\zeta(\mathbf{v}, I)$, the weak form evaluates as:

$$\int Q(f,g)(\mathbf{v},I)\zeta(\mathbf{v},I)\,dI\,d\mathbf{v} = \int \frac{f(\mathbf{v},I)g(\mathbf{w},J)}{(IJ)^\nu}\Big(\zeta(\mathbf{v}',I') - \zeta(\mathbf{v},I)\Big)dA. \tag{12}$$

The invariant measure dA natively incorporates the pre-collision density of states $(IJ)^\nu$:

$$dA = B\,\varphi_\nu(r)\psi_\nu(R)(1-R)R^{1/2}(IJ)^\nu\,dR\,dr\,d\hat{\mathbf{u}}'\,dJ\,d\mathbf{w}\,dI\,d\mathbf{v}. \tag{13}$$

Substituting dA analytically cancels the asymmetric quotient $(IJ)^\nu$ in the denominator. Applying the Galerkin projection—where the test and trial bases absorb the continuous equilibrium reference weights—the operator naturally splits into independent positive gain and negative loss components without artificial algebraic singularities.

We define the integrated inner polyatomic block, $\Omega^{(c)}_{poly}(z,x,y)$, which evaluates these split scattering cross-sections and internal energy transitions for a specific macroscopic interaction channel c. Factoring the polar deflection angle $\cos\chi$ outside the inner integrations yields:

$$\Omega^{(c)}_{poly}(z,x,y) = \int_{-1}^{1} d(\cos\chi)\Big[\mathcal{I}^{(c)}_{gain}(z,x,y,\cos\chi) - \mathcal{I}^{(c)}_{loss}(z,x,y,\cos\chi)\Big]. \tag{14}$$

3.1 The Loss Term Integration

For the loss term, the test functions evaluate the pre-collision states $v(z,x)$ and I. Because these are independent of the scattering angles and partition parameters, the azimuthal integral evaluates to 2π, and the geometric reconstructions are bypassed:

$$\begin{aligned}\mathcal{I}^{(c)}_{loss}(z,x,y,\cos\chi) &= 2\pi P^{(c)}_{loss}(y)R_{k_1,l_1}(v(z,x)) \\ &\quad\times \int_0^\infty dI\int_0^\infty dJ\int_0^1 dR\int_0^1 dr\,W_{4D}\,B(\dots)H_{i_1}(I)H_{i_2}(I)H_{i_3}(J),\end{aligned} \tag{15}$$

where W_{4D} encapsulates the standard equilibrium reference weights and partition functions.

3.2 The Gain Term Integration

The gain term requires evaluating the post-collision states v' and I'. To avoid repeated computation of the 3D spherical geometry, we apply Fubini's theorem, nesting the integrals based on their functional dependencies:

1. **Step 1: The Innermost r Integral.** Because v' and $\hat{\mathbf{v}}'$ do not depend on r, the radial polynomials and geometric Wigner filters factor outside this integral.

$$M(z, x, y, \cos\chi, I, J, R) = \int_0^1 dr\, W(r)\, B(\dots)\, H_{i_1}(I'(\dots)). \tag{16}$$

2. **Step 2: The Azimuthal ϵ Integral.** Independent of the r integral, we integrate the azimuthal angle ϵ. This encapsulates the macroscopic geometry and radial basis evaluation.

$$S^{(c)}(z, x, y, \cos\chi, I, J, R) = \int_0^{2\pi} d\epsilon\, R_{k_1,l_1}(v'(\dots))\, P^{(c)}_{gain}(\hat{\mathbf{v}}'(\dots), y). \tag{17}$$

3. **Step 3: The Kinetic Partition R Integral.** We merge the physical energy transition (M) and the macroscopic spatial rotation ($S^{(c)}$) by integrating over their shared dependency, R.

$$N^{(c)}(z, x, y, \cos\chi, I, J) = \int_0^1 dR\, W(R)\, M(\dots)\, S^{(c)}(\dots). \tag{18}$$

4. **Step 4: The Incident Internal Energy J Integral.** We integrate over the incident particle's internal energy, coupling the result to the incident test basis.

$$O^{(c)}(z, x, y, \cos\chi, I) = \int_0^\infty dJ\, W(J)\, H_{i_3}(J)\, N^{(c)}(\dots). \tag{19}$$

5. **Step 5: The Target Internal Energy I Integral.** Finally, we integrate over the target internal energy, completing the gain evaluation.

$$\mathcal{I}^{(c)}_{gain}(z, x, y, \cos\chi) = \int_0^\infty dI\, W(I)\, H_{i_2}(I)\, O^{(c)}(\dots). \tag{20}$$

3.3 Resolving the Post-Collision Square Root Singularity

A numerical consideration arises from the definition of the post-collision relative speed, $u' = \sqrt{2RE}$. Because the total energy is $E = \frac{1}{2}u^2 + I + J$, evaluating u' introduces a square root dependency on the internal energies. If the physical integrand contained odd powers of u', this would generate an infinite derivative at the boundaries where $I \to 0$ or $J \to 0$, reducing the spectral convergence rates of the Generalized Gauss–Laguerre quadrature rules.

However, the continuous Fubini ordering ensures that this singularity is resolved before the energy integrals are evaluated. The square root enters the physical integrand solely through the reconstruction of the post-collision target velocity vector: $\mathbf{v}' = \mathbf{U} + \frac{1}{2}u'\hat{\mathbf{u}}'$.

When the spectral test basis (a polynomial in the Cartesian components of $\mathbf{v}'$) is expanded algebraically, it generates cross-terms containing both even and odd powers of the coupled term ($u'\hat{\mathbf{u}}'$). Because the integrals over the solid angle defined by $\cos\chi$ and ϵ are evaluated prior to the internal energy variables, we integrate over the symmetric scattering sphere first. By geometric parity, the integral of any odd power of the direction vector $\hat{\mathbf{u}}'$ over the sphere evaluates to zero.

Consequently, only even powers of the post-collision relative speed survive the angular integration. Since $(u')^{2n} = (2RE)^n$, the surviving terms are polynomials in E. Expanding E, the integrand is a polynomial in I, J, and R. The square root is resolved analytically, confirming that the internal energy manifolds are free of boundary singularities and that standard unmapped Generalized Gauss–Laguerre rules can achieve spectral convergence.

4 Integration with the Duffy Patches

With $\Omega^{(c)}_{poly}(z, x, y)$ resolved through the continuous inner integrations, it depends on the macroscopic spatial variables. It therefore integrates into the 5D spatial Duffy architecture, factoring similarly to the monatomic formulation:

Patch 1 Evaluation ($y = xh$): The radial spatial weight $\mathcal{K}^{(c)}_{rad}$ depends solely on x, pulling outside the inner h-loop.

$$\mathcal{R}^{(c)}_{patch1} = \int_0^\infty dz\, z^{\gamma/2} e^{-z} \int_0^1 dx\, \mathcal{K}^{(c)}_{rad}(z, x)\, x \left[\int_0^1 dh\, \Omega^{(c)}_{poly}(z, x, y = xh) \right]. \tag{21}$$

Patch 2 Evaluation ($x = yh$): The radial weight depends on x, trapping it inside the inner h-loop.

$$\mathcal{R}^{(c)}_{patch2} = \int_0^\infty dz\, z^{\gamma/2} e^{-z} \int_0^1 dy\, y \left[\int_0^1 dh\, \Omega^{(c)}_{poly}(z, x = yh, y) \times \mathcal{K}^{(c)}_{rad}(z, x = yh) \right]. \tag{22}$$

By expanding Fubini's theorem via explicit functional dependencies, we show that the $\mathcal{O}(K_{\max})$ evaluation of the radial polynomial within $S^{(c)}$ is decoupled and factored out of the innermost r integration loop, bounding the arithmetic complexity.